Syllabic frequencies reveal a strong cross-linguistic prevalence of the Sonority Sequencing Principle

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Sonority is a phonological feature that describes the perceptual salience of a given phoneme relative to neighbouring ones (Blevins, 1995; S. Parker, 2008; S. G. Parker, 2002; Selkirk, 1984). While the acoustic correlates of sonority are still under debate, some physical features like intensity have been shown to strongly correlate with sonority indexes across spoken languages (Fant et al., 2000; Jany et al., 2007; S. Parker, 2017; S. G. Parker, 2002; but see Albert & Nicenboim, 2022 for a different perspective). Phonemes can be hierarchically organised according to their sonority, with those at the lower end having lower sonority and those at the higher end having higher sonority (e.g., Obstruents < Nasals < Liquids < Glides < Vowels, Clements, 1990). This hierarchy is often represented as a numerical scale (e.g., `1` for lowest sonority, `9` for highest sonority, S. Parker, 2008). According to this hierarchy, some syllables are considered well-formed across languages, whereas others are marked as ill-formed. The Sonority Sequencing Principle (SSP) is a phonotactic rule that captures how sonority constrains the ordering of phonemes within a syllable. The SSP states that each syllable should display a unique peak of sonority at the nucleus, and that sonority should rise toward the nucleus and fall away from it (S. G. Parker, 2002; Selkirk, 1984). Syllables are classified as following or violating the SSP by examining the sequence of sonority values assigned to each of its phonemes. For instance, /blif/ has a sonority contour of `2-6-8-3` (`8` being the vowel /i/, nucleus of the syllable) and is therefore considered well-formed. On the other hand, /lbif/ (`6-2-8-3`) is considered ill-formed because the onset displays a sonority dip from the first to the second phoneme, violating the progressive increase in sonority that characterizes well-formed syllables (see Table 1 for more examples).

Table 1

Examples of SSP-following and SSP-violating syllables in the dataset. Sonority index contours are shown for each syllable numerically and graphically. For example, the syllable /'skul/ in Assamese shows a violation at onset, as sonority drops from /s/ (sonority index 3) to /k/ (sonority index 1) before rising again at nucleus /u/ (sonority index 8).

Language	Syllable	Original word	Sonority index	
SSP-following				
Faroese	/aɻ/	altar /ˈaɻ̩.taɹ/	9-6	
Finnish	/berk/	Berg /ˈberk/	2-9-6-1	
Kankanaey	/bal/	balla /balˈla/	2-9-6	
Moksha	/alʲ/	альняка /ˈalʲ.nʲa.kə/	9-6	
Sanskrit	/beː/	कुबेर /ku.beː.ɾɐ/	2-9	
SSP-violating				
Assamese	/skul/	স্কুল /skul/	3-1-8-6	
German	/amts/	Amtsträger /ˈamtsˌtʁeːgɐ/	9-5-1-3	
English	/biːdz/	beads /biːdz/	2-8-2-4	
Serbian-Croatian-Bosnian	/br̂ːzo/	brzo /br̂ːzo/	2-6-4-9	
Mon	/hma/	ဘျော /hma/	6-5-9	

Experimental evidence suggests that the SSP plays a pivotal role in language processing, both in spoken (Berent et al., 2007, 2011; Berent & Lennertz, 2010; Miozzo & Buchwald, 2013; S. Parker, 2017) and signed languages (Perlmutter, 1993; Sandler, 1993; Stone et al., 2018). Studies on newborns, infants, and adults show a consistent preference toward well-formed (SSP-following) syllables over ill-formed (SSP-violating)

ones, even when well-formed syllables are unattested in the participants' native language. For instance, English-native adults prefer /bn/ (SSP-following) as syllabic onset over /lf/ (SSP-violating), although both clusters are not present in English (Berent et al., 2011; Berent & Lennertz, 2010). Few-day-old newborns show different brain haemodynamic responses when listening to SSP-following or violating syllables (e.g., *bran* vs. *lbif*) suggesting that precursors of the SSP are already present at birth (García-Castro et al., 2026; Gomez et al., 2014). Considering these findings, one would expect SSP-following structures to be more frequent than SSP-violating ones across virtually all languages, but a quantitative estimate of cross-linguistic prevalence of SSP-following structures is still elusive.

Violations of the SSP are not uncommon across languages. Examples include fricative-stop consonant clusters at syllable onset (e.g., /ʃt/, as in the German word Stein) and their reverse counterparts, stop-fricative clusters, in coda (e.g., /ks/, as in the English word clicks) (Kawasaki-Fukumori et al., 1997; Morelli et al., 2003). Languages that allow considerably more complex consonant clusters at onset tend to show SSP violations more frequently. This is the case of Polish, which allows for clusters of up to three consonants at syllable onset (e.g., *chrząszcz* /ˈxʂɔ̃ʂt͡ʂ/ [beetle]) (Gussmann, 2007; Jarosz, 2017), or Georgian (e.g., მწვრთნელი /mt͡sʼvrtʰneli/ [trainer]) (Crouch et al., 2023). Another case of a language with frequent violations of the SSP is Modern Standard Arabic, in which only 51% of its attested consonant clusters conform to the SSP (Al Tamimi & Al Shboul, 2013). To our knowledge, only two studies have provided a quantitative cross-linguistic approach toward the prevalence of the SSP. Both studies considered type frequency as an index of the prevalence of SSP-following or -violating structures. Type frequencies capture the number of times that a given structure (e.g., a certain type of consonant cluster) appears in the lexicon of a given language, or in the

repertoire of allowed sequences of clusters in a given language. Using this measure, evidence of a preference for SSP-following consonant clusters was been reported across 43 languages with an overall 72% compliance with the SSP across 97 consonant clusters (Frisch, 2015). Likewise, type frequencies have been examined in a larger pool of languages (496 from 58 language families), and at least 32% of the consonant clusters in word-onset position were found to violate the SSP, compared with 30% in word-final position (Yin et al., 2023).

While type frequencies have provided useful information about languages' tolerance of SSP violations (or lack thereof), they do not provide a direct estimate of how often speakers of such languages encounter SSP violations in natural speech due to two limitations. First, type frequencies necessarily reflect how often a given structure is used in natural speech. Consider an imaginary language that allows for 20 consonant clusters in total, one of which is /st/. While /st/ represents a sizeable portion of the language's phonotactical repertoire (5%), how frequently speakers encounter /st/ depends on at least two factors: the number of words containing /st/, and the frequency of each of those words. If /st/ occurs in only a few words or only in low-frequency words, its actual prevalence in natural speech would be lower than suggested by the language's phonological repertoire. Second, most studies extract their data from phonological transcriptions of words extracted from lexical inventories of a given language, but such phonological transcriptions often do not provide syllabic boundaries (Yin et al., 2023). Consequently, it is not always possible to include word-medial consonant clusters in the datasets, leaving a substantial portion of syllabic structures uttered in natural speech unobserved with only word-initial and word-final syllables being analyzed.

In the present study, we aim to overcome these two pitfalls to provide a quantitative estimate of cross-linguistic SSP prevalence that more accurately reflects the frequency with which speakers encounter SSP-following and SSP-violating syllables in natural speech. To this end, we first took into account a measure of syllabic token frequency, defined as the number of times a given syllable is uttered in a given language for every million words in a given corpus (Cholin & Levelt, 2009; Duchon et al., 2013; Perea & Carreiras, 1998). Second, we included a large sample of syllabified phonological transcriptions for each language in our dataset, including word-initial, word-medial and word-final syllables. We then used a Bayesian modelling approach to estimate the prevalence of the SSP in this dataset which comprised 52,588 syllables from 42 languages from 8 distinct language macro-families. This approach yielded estimates of the proportion of SSP-following syllables in each language, adjusted for the relative frequency of each syllable in natural speech.

Methods

Raw data, reproducible code, and project documentation are available in the Open Science Framework (https://osf.io/yeqms) and GitHub (https://github.com/NeuroDevCo/ssp-corpus).

Data collection and preprocessing

Phonological transcriptions were scraped from word entries in the Wiktionary database. Wiktionary is an open-source, free database containing 10,050,751 lexical entries from 8,244 languages contributed by 4,463,976 users (as of June 10th, 2026), many of which include phonological transcriptions of words in IPA. The wikipron Python package (version 1.3.3, Lee et al., 2020) was used to scrape all available phonological transcriptions of words for all 7,859 languages included in the Glottolog database (Nordhoff & Hammarström, 2011). This resulted in approximately 1.7 million

phonological transcriptions across 900 languages. Languages with less than 100 words were excluded, resulting in a final dataset of 731,145 words from 192 languages.

Available lexical frequencies for each word in each language were extracted using the *wordfreq* Python package (Speer, 2022), which provides an interface for retrieving lexical frequency data for words in over 40 languages. When lexical frequencies were not available for a particular language (e.g., the ISO 639-3 code did not match any language available in *wordfreq*), the closest equivalent language was used, if available. Equivalent languages were identified based on their distance from the original language according to CLDR v38.11[1]. The rationale for using equivalent languages to retrieve missing lexical frequencies was that users of close languages are likely to use similar words to refer to the same objects, and lexical frequencies were therefore expected to be reasonably comparable across equivalent languages. Words with missing lexical frequency scores were excluded from the dataset.

The phonological transcriptions of words were then segmented into syllable candidates. When possible, syllable boundary markers were used, either explicit (".") or implicit ("ˈ" for primary stress, "ˌ" for secondary stress). In the absence of syllable markers, the word was assumed to be monosyllabic. Segments of each syllable candidate were then classified as consonants (C), vowels (V), or semi-vowels (v). Individual vowels, consecutive strings of vowels and semi-vowels (e.g., V, VV, Vv, VVv) were classified as vocalic clusters. Syllable candidates with more than one vocalic cluster (e.g., CVvCV, VCV) were flagged as multisyllabic and excluded from the dataset. This methods allowed to extract word-medial syllables when possible. Duplicated

[1]The CLDR system assigns a numeric value to each pair of language codes based on whether both codes are assigned to identical languages with different names (e.g., Norwegian Bokmål → Norwegian, score 0), minor regional differences (score 1-3, e.g., Norwegian Bokmål → Norwegian), and languages with significant but unproblematic regional differences (score 10-24, e.g., Afrikaans → Dutch, Wu Chinese → Mandarin Chinese), etc.

syllables within each language were removed (e.g., if the syllable /ka/ appeared in the Spanish words /ˈka.sa/ and /kaˈpo/, it was counted only once in the Spanish dataset). This procedure yielded one syllabic repertoire for each language.

To evaluate whether each extracted syllable followed or violated the SSP, a sonority score was assigned to each phoneme on a scale from `1` to `9`, using the *panphon* Python package (version 0.22.2, Mortensen et al., 2016). Syllables with more than one peak (i.e., exhibiting a local minimum in the sonority contour) were classified as SSP-violating, and as SSP-following otherwise. Following this rule, syllables with a flat sonority contour (e.g., English word sphinxs /ˈsfɪŋks/, `3-3-8-5-1-3`) were classified as SSP-following. Whether consecutive segments with same sonority index comply or not with the SSP is a matter of debate (e.g., Tropf, 1987; Berent et al., 2007; Eckman, 1987; Santolin et al., 2023). for this reason, we report complementary analyses in which flat sonority contours were classified as SSP violating, showing equivalent results (see Appendix B).

To calculate the frequency of each syllable, we summed the lexical frequency (counts per million words) of each word the syllable was part of in the dataset (see Duchon et al., 2013 for a similar approach). The rationale behind this approach is that each time a word is uttered, all of its syllables are uttered as well. Therefore, summing the lexical frequencies of all words containing a given syllable yields an estimate of that syllable's frequency in natural speech. The final dataset comprised 52,588 syllables from 42 languages from 8 distinct language macro-families.

Data analysis

We used a multilevel, Bayesian logistic regression model to estimate the probability of each syllable in the dataset following (or violating) the SSP, considering the language it belonged to, and its syllabic frequency. We modelled the response

variable $y_i j$ (`1`, if the syllable was SSP-following, `0` otherwise) using a Bernoulli probabilistic distribution with probability parameter θ_{ij}. We specified a linear regression model for θ_i, which included fixed regression coefficients for the intercept (α) and the slope of syllabic frequency (β), and random effects u_j and w_j for each fixed coefficient, respectively. We specified weakly informative priors for all parameters in the model (see Equation 1). To estimate the posterior distribution of the model we used Hamiltonian Monte Carlo as implemented in *PyMC3* (version 6.2.0, Patil et al. (2010)), with four chains of 2,000 iterations, 1,000 of which were warm-up iterations.

Likelihood:

Global prevalence of the SSP:

$$
\begin{aligned}
y_{ij} &\sim \text{Bernoulli}\ (\frac{\theta_{ij}}{1-\theta_{ij}}) \\
\theta_{ij} &= \alpha + u_j + (\beta + w_j) \cdot \text{Frequency}_{ij}
\end{aligned}
$$

Prior:

$$
\begin{aligned}
\alpha, \beta &\sim \text{Normal}\ (0, 1) \\
u_j, w_j &\sim \text{Normal}\ (0, \sigma_{u,w}) \\
\sigma_{u,w} &\sim \text{HalfNormal}\ (0.5)
\end{aligned}
\tag{1}
$$

Where:

i indexes each observation and j indexes the language it belongs to

y_{ij} indexes whether the syllable follows (1) or violates (0) the SSP

u_j, w_j are language-specific adjustments to α and β

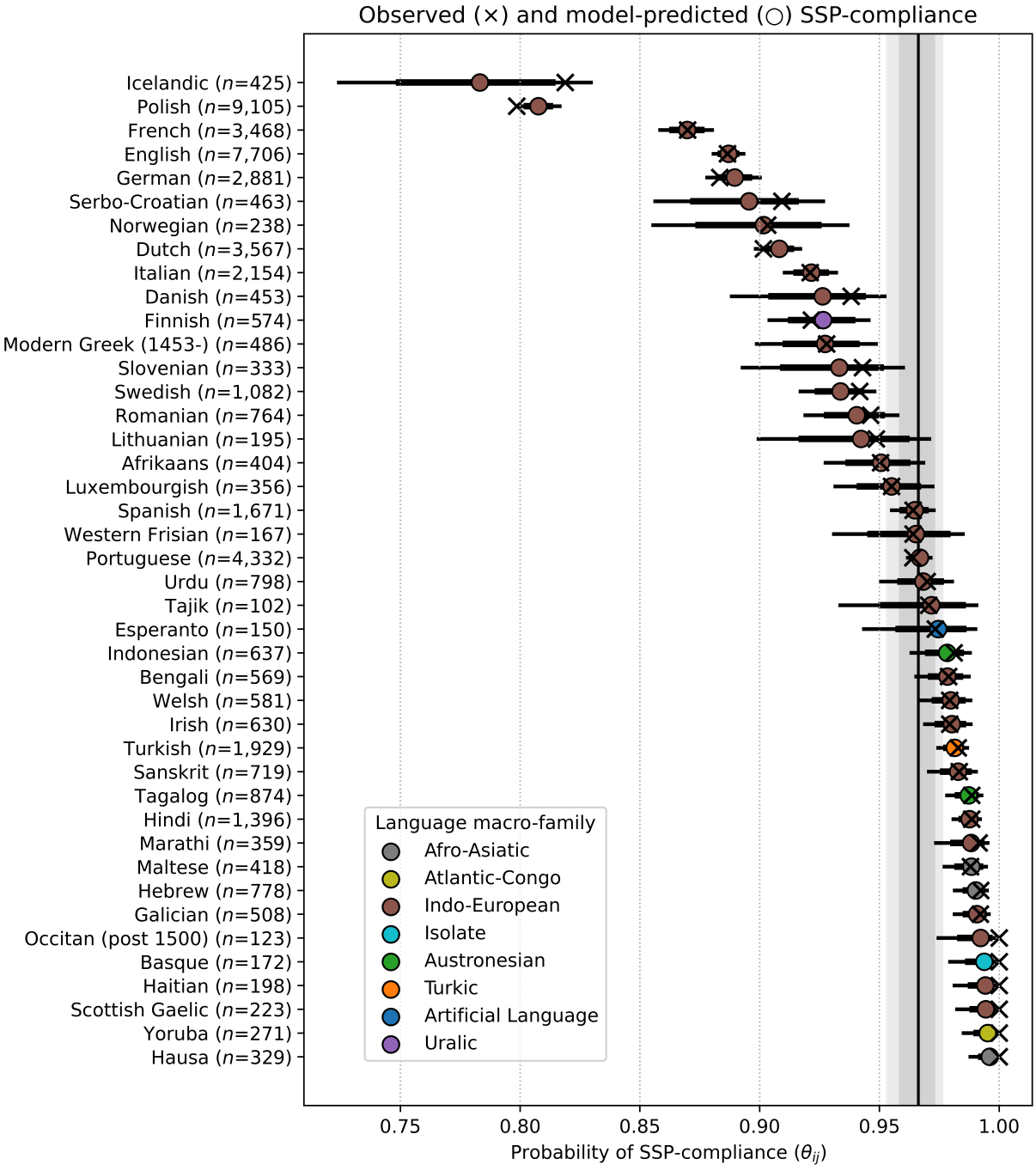
Results

We report the median and 95% credible interval (CrI) of the posterior distribution of the model parameters of interest. The 95% CrI contains the 95% of the posterior distribution of the parameter. This interval can be interpreted as being 95% likely to contain the true value of the parameter, given the observed data and the prior distribution of the model (Kruschke & Liddell, 2018).

The global proportion of SSP-following syllables across languages was θ = 0.966 (95% CrI = [0.953, 0.977]), and the language-specific prevalence of the SSP ranged from 78.319% (Icelandic, 95% CrI = [72.353%, 81.730%]) to 99.605% (Hausa, 95% CrI = [98.720%, 99.918%]) (see Figure 1). Only 2.38% of the languages in the sample showed less than 80% of SSP prevalence, 14.29% languages showed less than 90% of SSP prevalence, and 38.10% languages showed less than 95% of SSP prevalence.

Figure 1

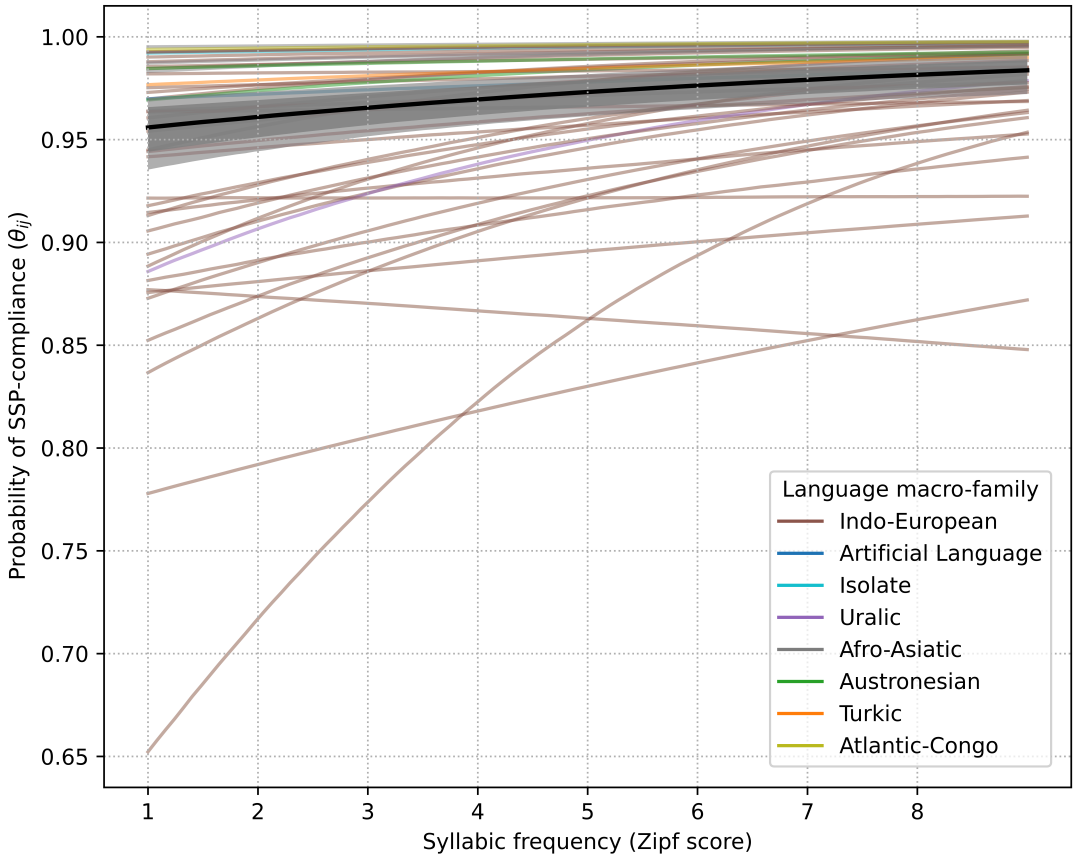
Observed (crosses) and model-predicted (circles) estimates of SSP prevalence across languages in the dataset. Intervals indicate the 95% and 79% CrI of the posterior distribution. The number of syllables included in the dataset is indicated next to each language.



Syllabic frequency showed a positive, non-zero association with the probability of SSP compliance (β = 0.151, 95% CrI = [0.064, 0.246]), indicating that SSP-following syllables were, on average, more frequent than SSP-violating syllables. For every standard deviation increment in syllabic frequency (corresponding to 1.19 Zipf scores), the probability of SSP compliance increased a maximum of 0.0378 probability points (95% CrI = [0.016, 0.0614]) (see Figure 2).

Figure 2

Mean predictions of SSP compliance as a function of syllabic frequency. Predicted cross-language probability of SSP compliance (θ_{ij}) is showed as a black line. Thin lines indicate language-specific predictions, color-coded by language macro-family.



Discussion

The present study was aimed at quantifying the cross-linguistic prevalence of the Sonority Sequencing Principle (SSP). Previous experimental studies had pointed to the SSP as a possible universal constraint to syllabic structure that guides language processing (e.g., Berent & Lennertz, 2010; Berent et al., 2011; Gomez et al., 2014), but whether such contraint was reflected in an overall quantitative preference for SSP-following structures was unclear. Capitalising on syllable token frequencies, we used Bayesian modelling to estimate to average probability of SSP conformity in a reportoire of 52,588 syllables from 42 languages from 8 distinct language macro-families. We report an overall cross-linguistic 96.63% comformity with the SSP, and more than 85.71% of the languages in the sample showing more than 90% comformity.

The cross-linguistic prevalence of the SSP here reported exceeds estimates from prior studies, which relied on type frequencies of consonant clusters to derive such measure. For instance, Frisch (2015) reported that 72% of attested consonant clusters across 47 languages followed the SSP. Other studies on individual languages reported similar quantities, like San'ani Arabic (Watson, 2002) or Modern Standard Hebrew (Gishri, 2009), while in other cases like Modern Standard Arabic, SSP-following consonant clusters were reported to represent around only 51% of allowed consonant clusters Al Tamimi and Al Shboul (2013). Contrary to prior work on the prevalence of the SSP, the present study used token frequencies (as opposed to type frequencies) to adjust the estimated prevalence of SSP-following structures, therefore incorporating the frequency of use in natural speech of each of the syllables present in the dataset in our analysis. The higher prevalence of SSP compliance here reported might be the result of type frequencies underestimating how often SSP-following consonant clusters occur in natural speech. In support of this possibility, we found a positive, non-zero association

between token syllabic frequency and the probability of SSP compliance: syllables that occur more often in natural speech show a stronger tendency to follow the SSP than those that occur less often, indicating an asymmetric frequency of use between SSP-following and SSP-violating structures, which type frequencies do not capture.

These results suggest that violations of the SSP are exceptional for the majority of the languages included in our dataset. Though, conclusions about the universality of the SSP should taken with caution. First, violations of the SSP, while uncommon, are pervasive in virtually all languages considered in this study, as well as those not here considered by it (Yin et al., 2023). As a consequence, the SSP cannot be taken as a hard linguistic universal that is always complied with, but rather as an overall preference toward certain linguistic structures that that virtually all languages display to a varying extent, which may interact with other—perhaps language-specific—contraints to syllabic structure (Prince & Smolensky, 2004). This approach may provide a better understanting of the role of the SSP in syllabification. An example is the edge case of Imdlawn Tashlhiyt dialect of Berber, in which despite obstruent voiceless consonants (with minimum sonority) being allowed as vocalic nuclei, sonority has proven useful to generate predictions about syllabic well-formedness (Dell & Elmedlaoui, 1985; Prince & Smolensky, 2004).

Our findings have relevant implications for language acquisition research, as it provides further evidence that a remarkably large proportion of the syllables encountered by humans in their first months of age may follow SSP contraints. Syllables are a privileged linguistic unit in speech processing during the earliest stages of language acquisition (Bertoncini et al., 1988; Eimas, 1999; Jusczyk & Derrah, 1987). During as early as their first days of life humans are able to discriminate between word-forms based on the number of syllables, instead of relying on the number of phonemes

or the duration of the produced utterance (Bijeljac-Babic et al., 1993). Newborns can also track the co-ocurrence of syllable-sized groups of speech sounds to chunk the continuous speech stream into word-like units (Fló et al., 2019, 2022), an ability that will later give them access to the precursors of a lexicon (Saffran et al., 1996; Saffran & Kirkham, 2018). Given the relatively limited experience newborns have accumulated with their native language, their ability to identify syllables in speech is remarkable. Newborns' and young infants' have been found to be sensitive to certain contraints to syllabic structure (e.g. Bertoncini & Mehler, 1981; Moon & Fifer, 1990; Santolin et al., 2024) and—more critically for this study—to the SSP (García-Castro et al., 2026; Gomez et al., 2014). In the context of our findings, such an early knowledge about the SSP would provide humans with the ability to correctly segment most of the syllables they encounter in the speech stream, acting as a precursor for later, more complex computations involved in language acquisition.

Several points need to be considered when interpreting the findings in the present study. First, as suggested by prior work, relying on non-exhaustive databases in which datapoints are selected for inclusion in non-systematic ways, like the case of Wiktionary, may lead to a lower representativity of the dataset (Yin et al., 2023). Each syllable in our dataset was included based on the availability of at least one word containing that syllable in its corresponding language in Wiktionary. Word entries in Wiktionary are added by volunteers and mantainers, and the mechanisms behind which words are included or missing in Wiktionary, and how much information is included in the entry of each word, are unknown and most likely non-random. For instance, it is possible that words with higher lexical frequency (words used more often by the speakers of a given langugae) are more likely to be present in Wiktionary, or more likely

to include a phonological transcription, compared to less frequent words. Because of this, our dataset may be, to some extent, subject to selection bias.

Second, while the present study includes multiple languages from a range of different language macro-families, the number of observations gathered for some languages is reduced, especially when comparing Indo-European languages (clearly overrepresented in our dataset) with languages from other macro-families. This unequal availabity of data across languages is partially accounted for by the modelling approach adopted. The estimation of language-specific parameters was informed by the estimation of global parameters, as a result of the partial pooling of random effects that multilevel models implement (Gelman et al., 2007). This means that in languages with fewer observations, extreme scores were less likely to be biases the estimations of that and other languages than if data had been modelled separately for each language, or altogether ignoring cross-language variability. In addition, the Bayesian credible intervals reported provide a more transparent account of the uncertainty around its generated estimates compared to classical (frequentist) confidence intervals, as they can be interpreted as containing the true value of its corresponding parameter with some degree of confidence (i.e., 95%), given the model, data, and priors. As a result, language-specific estimates reported for languages with fewer observations are acompanied more uncertainty, preventing overconfident interpretations of the model outputs for such languages.

Finally, it should be considered that the phonological transcriptions included in the dataset are the result of how one or more Wiktionary volunteers decided to transcribe a given word. Because phonological transcriptions are an abstract representation of the phonology of a string of sounds in a specific language, they do not always caputre all relevant features of the acoustic or, more generally, the physical form

of the word being transcribe. For instance, epenthetic vowels may or may not be included in a phonological transcription enclosed in brackets for phonetic detail. The decision of including this symbol or not falls on the volunteer to ultimately enters the phonological trancription in the database, which may not represent what a larger group of linguists would decide, perhaps because of linguistic disputes about phonological notation, or because several transcriptions of the same word are permitted due to dialectal variations.

Overall, the present study updates prior estimates of cross-linguistic compliance with the SSP, pointing to both a substantially higher presence of SSP-following syllables than SSP-violating syllables across languages, and to the former syllables ocurring more frequently in natural speech than the latter.

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Appendix A

Number of SSP-following and SSP-violating syllables per language

Table A1

Number of SSP-following and SSP-violating syllables for each language in the dataset.

ISO-639-3	Words	Syllables	SSP+	%	ISO-639-3	Words	Syllables	SSP+	%
yor	1,311	271	271	100.0%	spa	106,961	1,671	1,611	96.4%
eus	10,555	172	172	100.0%	fry	425	167	161	96.4%
oci	243	123	123	100.0%	por	99,468	4,332	4,176	96.4%
hau	738	329	329	100.0%	ltz	860	356	340	95.5%
hat	848	198	198	100.0%	afr	619	404	384	95.0%
gla	1,112	223	223	100.0%	lit	279	195	185	94.9%
heb	948	778	772	99.2%	ron	2,126	764	723	94.6%
glg	3,477	508	504	99.2%	slv	1,704	333	314	94.3%
mar	703	359	356	99.2%	swe	1,624	1,082	1,019	94.2%
tgl	13,663	874	864	98.9%	dan	1,399	453	425	93.8%
hin	5,072	1,396	1,380	98.9%	ell	3,104	486	451	92.8%
mlt	3,777	418	413	98.8%	fin	73,343	574	529	92.2%
san	2,831	719	707	98.3%	ita	15,211	2,154	1,984	92.1%
tur	5,108	1,929	1,896	98.3%	hbs	8,822	463	421	90.9%
ind	3,851	637	625	98.1%	nor	451	238	215	90.3%
gle	4,682	630	617	97.9%	nld	11,340	3,567	3,216	90.2%
cym	5,150	581	569	97.9%	eng	36,929	7,706	6,832	88.7%
ben	2,771	569	557	97.9%	deu	13,142	2,881	2,545	88.3%
epo	5,975	150	146	97.3%	fra	17,659	3,468	3,017	87.0%
tgk	258	102	99	97.1%	isl	2,346	425	348	81.9%
urd	1,088	798	774	97.0%	pol	123,564	9,105	7,271	79.9%

Appendix B

Results after classifying flat sonority contours as SSP-violating

Figure B1

Observed (crosses) and model-predicted (circles) estimates of SSP prevalence across languages in the dataset. Intervals indicate the 95% and 79% CrI of the posterior distribution. The number of syllables included in the dataset is indicated next to each language.

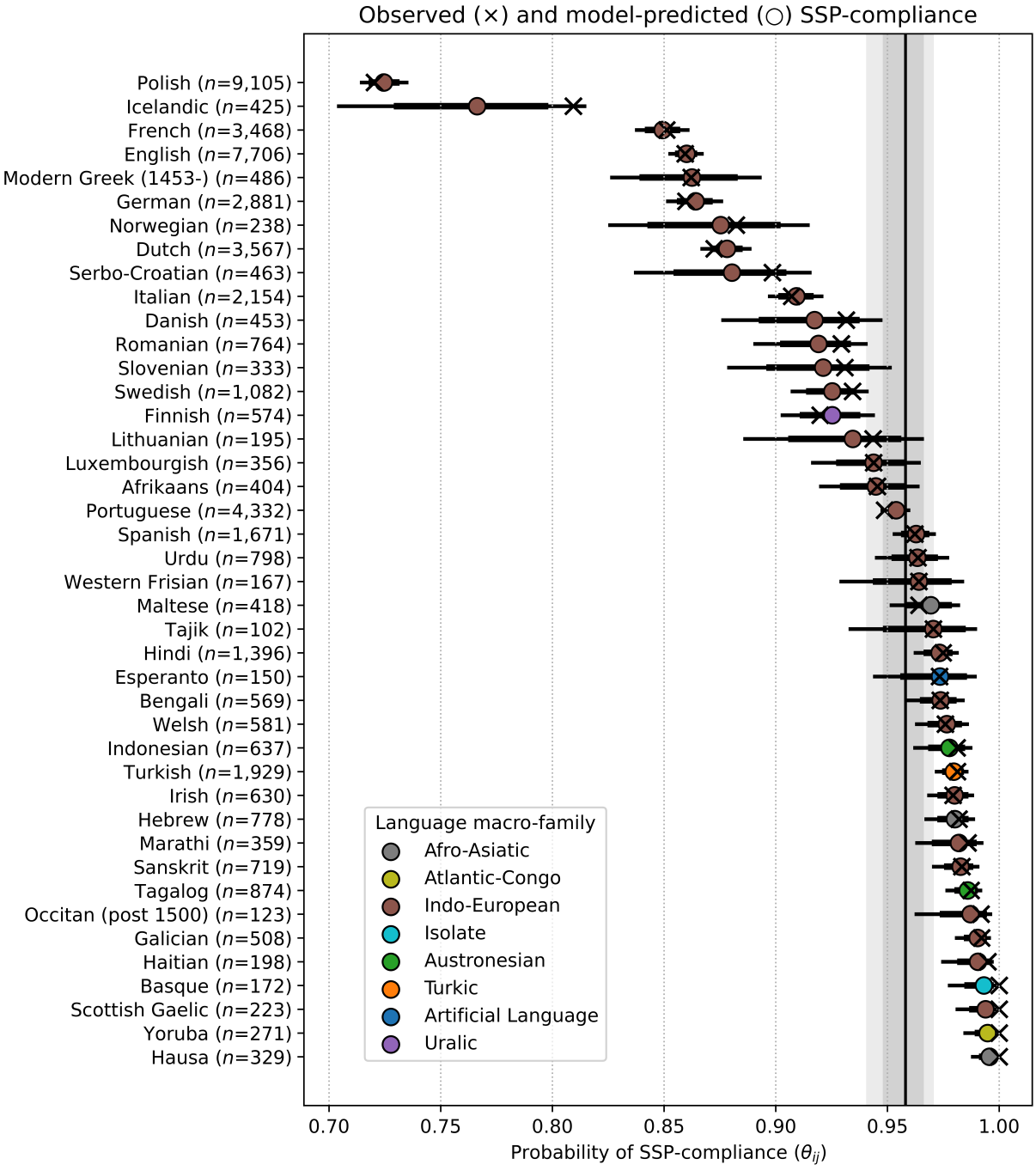


Figure B2

Mean predictions of SSP compliance as a function of syllabic frequency. Predicted cross-language probability of SSP compliance ($\theta_i j$) is showed as a black line. Thin lines indicate language-specific predictions, color-coded by language macro-family.

